James R. Bounds, PhD

Curriculum Vitae

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EXECUTIVE SUMMARY

Experimental Physicist (PhD) and Hardware/Software Engineer with 10+ years of experience designing precision optical instruments, custom analog electronics, and scientific automation software. Expert in laser spectroscopy, frequency combs, and high-performance numerical computing.

EDUCATION & EMPLOYMENT HISTORY

2015 — **B.S. Physics**
Texas A&M University
Department of Physics and Astronomy

2012–2015 — **Undergraduate Research Assistant**
Texas A&M University
Advisor: Prof. Hans Schuessler

2015-2016 — **IT Technician**
Systek Computing
College Station, Texas

2016–2025 — **Graduate Research Assistant**
Texas A&M University
Advisor: Prof. Hans Schuessler

2016-2025 — **Visiting Scientist (Summer terms)**
Max Planck Institute of Quantum Optics
Supervisor: Prof. Thomas Udem

2025–2026 — **Graduate Teaching Assistant**
Texas A&M University
Advisor: Prof. Hans Schuessler, Prof. Alexei Sokolov

2026 — **PhD. Physics**
Texas A&M University
Advisor: Prof. Alexei Sokolov

RESEARCH EXPERIENCE

- **Designed, assembled, and aligned high-precision optical setups:**
  Including Cavity Ring-down Spectroscopy (CRDS), Vernier spectroscopy, and Doppler-free saturated absorption systems. These systems achieved sub-ppb sensitivity in trace-gas detection.
- **Theoretical modeling and optimization of complex optical systems:**
  Built Gaussian beam simulation framework and custom ray/wave propagation tools in Python and C to validate beam geometry and cavity coupling efficiency.
- **Engineered custom optoelectronic and RF detection hardware:**
  Full life-cycle development of custom high-bandwidth amplified photodetectors, analog signal conditioning circuits, and mixed digital/analog devices for the optimization of experimental signals. This development spans the conception, schematic, KiCAD PCB layout, assembly, and enclosure design.
- **Calibrated and characterized mid-IR and near-IR laser systems:**
  Used optical frequency combs, custom-built wavemeters and interferometers for ultra-high-resolution wavelength metrology, optical frequency metrology, and hyperfine structure determinations.
- **Developed full-stack laboratory automation and data acquisition (DAQ) software:**
  Development spans from hardware-level FPGA fabric design for accelerated real-time processing, OS-level drivers written in C, to the high level interface written in Python and QT.
- **Constructed precision optomechanical assemblies and vacuum interface systems:**
  utilizing CAD software (Autodesk Inventor), 3D printing, and hands-on lathe/milling fabrication to ensure structural stability and low thermal drift of optical and vacuum-compatible mounts.
- **Executed optical fiber characterization and assembly:**
  including single-mode and polarization-maintaining fiber splicing, free-space-to-fiber coupling, and loss minimization across visible and mid-IR wavelengths.

- **Led end-to-end experimental error analysis and validation:**
  Processing raw spectral measurements against line-list databases (eg. HITRAN) using custom non-linear fitting and numerical optimization algorithms, in addition to error quantification.

- **Collaborated across international research divisions:**
  (including Max Planck Institute of Quantum Optics), delivering hardware prototypes and presentation deliverables on tight timelines.

PROJECTS AND HARDWARE SYSTEMS

- **Optical Breath Analysis Platforms:** Complete design, fabrication, assembly and alignment of both cavity-ringdown spectrometers and Vernier spectrometers. Both systems are easily capable of sub-1ppb trace gas detection in human breath samples.

- **Wavelength Metrology Platform:** Construction of a real-time wavelength analysis instrument capable of precision of less than 1 part in 10^8, aided by two complementary algorithms invented in my dissertation for the analysis of instrument data.

- **Ultra-high precision IR spectrometer:** Conception, design, and implementation of a saturated absorption spectrometer with on-board optical frequency calibration using a comb. This was used to measure the hyperfine spectrum of metastable xenon with accuracy competitive with much more complex systems.

PUBLICATIONS

Peer Reviewed

2025 **Acetone Absorption Cross-Section in the Near-Infrared of the Methyl Stretch Overtone and Application for Analysis of Human Breath**.
Bounds, J., E. Aluauee, A. Kolomenskii, and H. Schuessler.
Optics **6** (1)
, URL: https://www.mdpi.com/2673-3269/6/1/9.

2023 **Hyperfine structure and isotope shifts of xenon measured for near-infrared transitions with Doppler-free saturated absorption spectroscopy**.
Bounds, J., A. Kolomenskii, R. Trainham, M. Manard, et al.
Spectrochimica Acta Part B: Atomic Spectroscopy **202**, p. 106635
URL: https://www.sciencedirect.com/science/article/pii/S0584854723000228.

2022 **Spectroscopic trace gas detection in air-based gas mixtures: Some methods and applications for breath analysis and environmental monitoring**.
Xia, J., F. Zhu, J. Bounds, E. Aluauee, et al.
Journal of Applied Physics **131** (22), p. 220901
URL: https://doi.org/10.1063/5.0091263.

2019 **Sensitive acetone detection with a mid-IR interband cascade laser and wavelength modulation spectroscopy**.
Xia, J., F. Zhu, A. A. Kolomenskii, J. Bounds, et al.
OSA Continuum **2** (3), pp. 640–654
URL: https://opg.optica.org/osac/abstract.cfm?URI=osac-2-3-640.

2018 **Sensitive Spectroscopic Analysis of Biomarkers in Exhaled Breath**.
Bicer, A., J. Bounds, F. Zhu, A. A. Kolomenskii, et al.
International Journal of Thermophysics **39** (6), p. 69
URL: https://doi.org/10.1007/s10765-018-2389-9.

2017 **Single ion fluorescence excited with a single mode of an UV frequency comb**.
Ozawa, A., J. Davila-Rodriguez, J. R. Bounds, H. A. Schuessler, et al.
Nature Communications **8** (1), p. 44
URL: https://doi.org/10.1038/s41467-017-00067-9.

2017 **Probing methane in air with a midinfrared frequency comb source**.
Zhu, F., J. Xia, A. Bicer, J. Bounds, et al.
Appl. Opt. **56** (22), pp. 6311–6316
URL: https://opg.optica.org/ao/abstract.cfm?URI=ao-56-22-6311.

2015 **Mid-infrared dual frequency comb spectroscopy based on fiber lasers for the detection of methane in ambient air**.
Zhu, F., A. Bicer, R. Askar, J. Bounds, et al.
Laser Physics Letters **12** (9), p. 095701
URL: https://doi.org/10.1088/1612-2011/12/9/095701.

2014 **Near infrared frequency comb vernier spectrometer for broadband trace gas detection**.
Zhu, F., J. Bounds, A. Bicer, J. Strohaber, et al.
Opt. Express **22** (19), pp. 23026–23033
URL: https://opg.optica.org/oe/abstract.cfm?URI=oe-22-19-23026.

Preprint

2026 ***Carrier Revival in Long Trapped-Ion Chains***.
Egli, F., C. Shanks, J. Bounds, J. Moreno, et al.
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CONFERENCE PROCEEDINGS

2023 **Hyperfine Structure of Xe by Frequency Comb Calibrated Saturation and Collinear Fast Beam Spectroscopies**
31st Annual International workshop, Laser Physics 2023
H.Schuessler, J. Bounds, N. Schnoor, A. Kolomenskii, R. Trainham, M. Manard.

2023 **Local and Remote Measurements of Atmospheric Trace Gases**
31st Annual International workshop, Laser Physics 2023
A. Kolomenskii, C. Rodriguez, J. Bounds, E. Aluauee, H. Schuessler.

2021 **Broadband Frequency Comb Vernier Spectroscopy In The Mid-IR With Video-recording Retrieval Of Absorption Spectrum**
International Symposium on Molecular Spectroscopy
H. Schuessler, J. Bounds, A. Kolomenskii
DOI: 10.15278/isms.2021.FF12

2021 **Spectroscopic Detection Of Methane at a ppt Sensitivity Level in Mid-IR with a Long-path Multipass Cell**
International Symposium on Molecular Spectroscopy
J. Xia, C. Rodriguez, J. Bounds, A. Kolomenskii, H. Schuessler
DOI: 10.15278/isms.2021.RF10

2018 **Frequency comb vernier spectroscopy of methane in the mid-IR with temporal retrieval of comb lines**
International Symposium on Molecular Spectroscopy
James R Bounds, F. Zhu, A. Kolomenskii, H. Schuessler.
DOI: 10.15278/isms.2018.WA05

2018 **Acetone and methane detection with wavelength modulation spectroscopy in the near- and mid-IR**
International Symposium on Molecular Spectroscopy
J. Xia, F. Zhu, J. Bounds, A. Kolomenskii, H. Schuessler.
DOI: 10.15278/isms.2018.RH03

POSTER PRESENTATIONS

2025 **Hyperfine structure and isotope shifts of xenon measured for near-infrared transitions with Doppler-free saturated absorption spectroscopy**
SIBOR Workshop
Cook's Branch Conservancy, Montgomery, Texas

ORAL PRESENTATIONS

2024 **Frequency Comb Calibration of the Hyperfine Structure of Metastable Xenon**
SIBOR Workshop
Cook's Branch Conservancy, Montgomery, Texas

2023 **Frequency Comb Calibration of the Hyperfine Structure of Metastable Xenon**
SIBOR Workshop
Cook's Branch Conservancy, Montgomery, Texas

2022 **Metastable Xenon Hyperfine Structure Measurement**
SIBOR Workshop
Cook's Branch Conservancy, Montgomery, Texas

2019 **Vernier Spectroscopy and Spectroscopic Detection of Biomarkers in Exhaled Breath**
Workshop of Max Planck Institute of Quantum Optics Laser Spectroscopy Division
Schloss Ringberg, Kreuth, Germany

OPEN SOURCE PROJECTS

Github: https://github.com/jaymz07

- **Battery-Powered, Amplified Photodetector**
  https://github.com/jaymz07/Photodetector-Battery
- **Quantum Tunneling Demo Written in Java**
  https://github.com/jaymz07/quantum-tunneling
- **Gaussian Beam Simulator GUI**
  https://github.com/jaymz07/Gaussian-Beam-Sim-Qt
- **Grotrian Diagram Generator**
  https://github.com/jaymz07/Grotrian

TECHNICAL SKILLS

Optics & Photonics: Precision Free-Space Alignment, Optical Frequency Combs, Cavity-Enhanced Spectroscopy (CRDS, Vernier), Dual-Comb Spectroscopy, Optical Fiber Splicing & Coupling, Interferometry, Wavelength Metrology, Multi-Pass Cells, Vacuum Optical Interfaces.

Hardware, EDA & Electronics: Analog/Digital/Mixed-Circuit Design, Schematic Capture & PCB Layout (KiCAD), Circuit Simulation (NI Multisim, QUCS), FPGA Fabric Design (Verilog), Microcontrollers & Embedded Systems, Photodetector Optimization, Signal Conditioning, PID Controllers, VNAs, Oscilloscopes.

Instrumentation & DAQ: Hardware Protocol Automation (GPIB, RS232, USB, Ethernet, SPI, I^2C, PyVISA), Custom DAQ Systems, FPGA Acceleration, GUI Development (Qt/PyQt, Tkinter, LabVIEW).

Programming Languages: Python, C, C++, Cython, Verilog, MATLAB, Java, Bash, Shell Scripting, Git, LaTeX.

Scientific Computing & Modeling: NumPy, SciPy, PyTorch, Numba, OpenCV, Scikit-learn, Pandas, Slurm HPC Cluster Management, Ray-Tracing & Gaussian Beam Modeling.

Mechanical: 3D and CAD Modeling (Autodesk Inventor, Blender, OpenGL, PyVista), Precision Machining (Vertical Mill, Precision Lathe), 3D Printing / Additive Manufacturing, Ultra-High Vacuum (UHV) Systems.

Systems & Infrastructure: Linux Systems Administration (openSUSE, Debian, Arch, Fedora, RHEL/CentOS), SSH Server Configuration, Network Infrastructure, X11 Forwarding.